Laboratório 6: ULA com decodificador para display de 7 segmentos

INTEGRANTES:

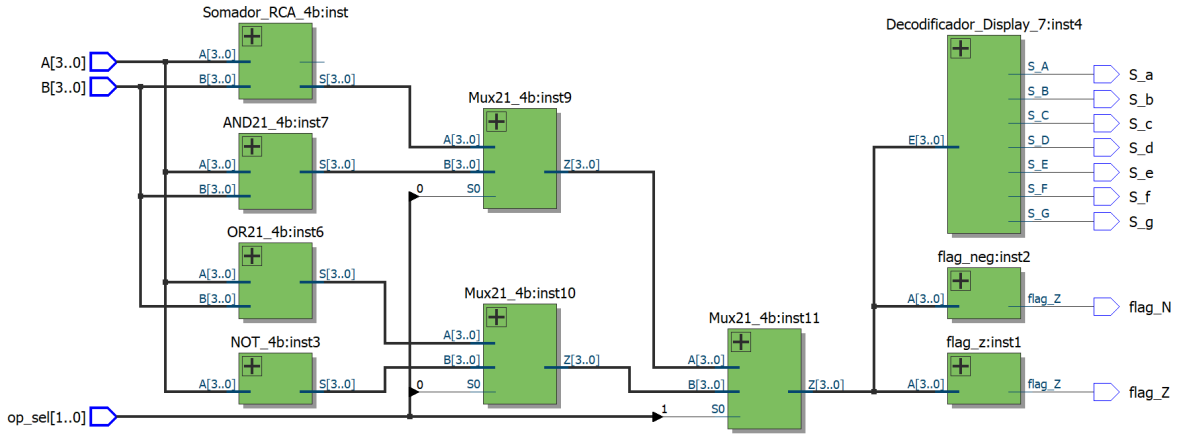
Nicolas Fagundes Mendonça - 00344411

Henrique Vieira Alves - 00244439

Objetivo:
Projetar e simular um decodificador de binário para a representação hexadecimal em display de
7 segmentos. Após, programar na FPGA a ULA do LAB05 de 4 bits com a saída desse circuito, para
poder visualizar o resultado no display da placa Cyclone II.

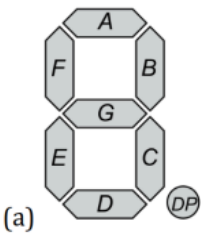
O circuito implementado neste laboratório é uma ULA com Display Funcional em placa.

ULA COM DISPLAY:

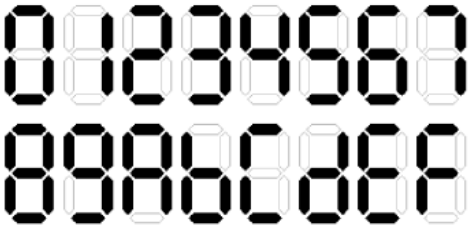


Possui:

- 3 Entradas. Sendo elas a Entrada A e B para implementar números de 4 Bits, e uma entrada de seletor de Operação. Ambas entradas A e B possuem 4 Pinos para escolher números de 1 a 8. A entrada de Operação possui 2 Pinos para escolher as operações de SOMA, NOT, ANDou OR
- 9 Saídas. Para representar as FLAGS Z ou N e as Saídas do Display (A,B,C,D,E,F & G) conforme figura abaixo:



(a)



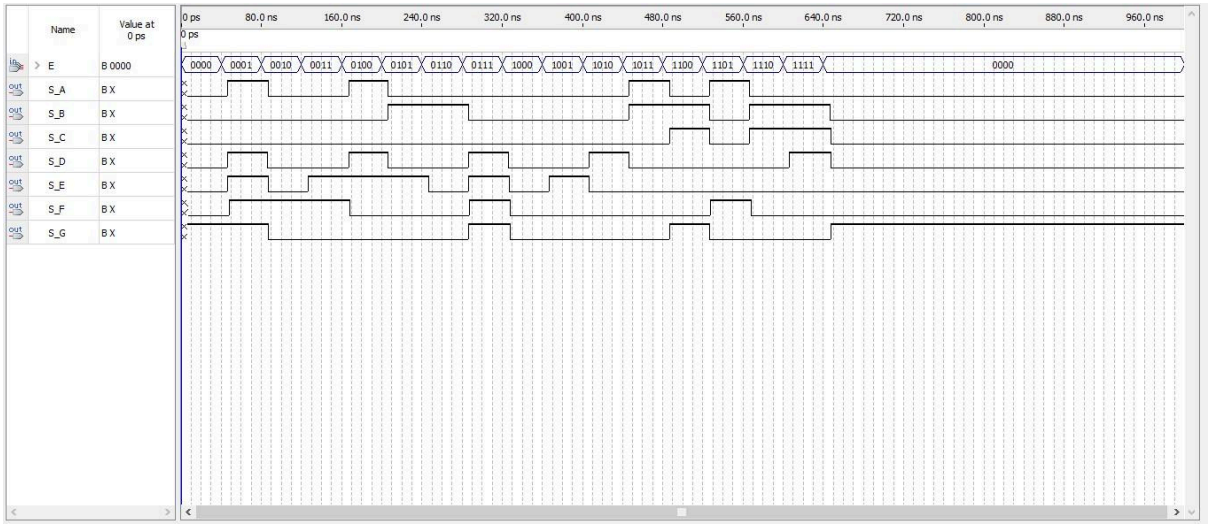
(b)

Flow Summary	
Flow Status	Successful - Thu May 15 19:50:04 2025
Quartus II 64-Bit Version	13.1.0 Build 162 10/23/2013 SJ Web Edition
Revision Name	ULA
Top-level Entity Name	ULA
Family	Cyclone III
Device	EP3C16F484C6
Timing Models	Final
Total logic elements	18
Total combinational functions	18
Dedicated logic registers	0
Total registers	0
Total pins	19
Total virtual pins	0
Total memory bits	0
Embedded Multiplier 9-bit elements	0
Total PLLs	0

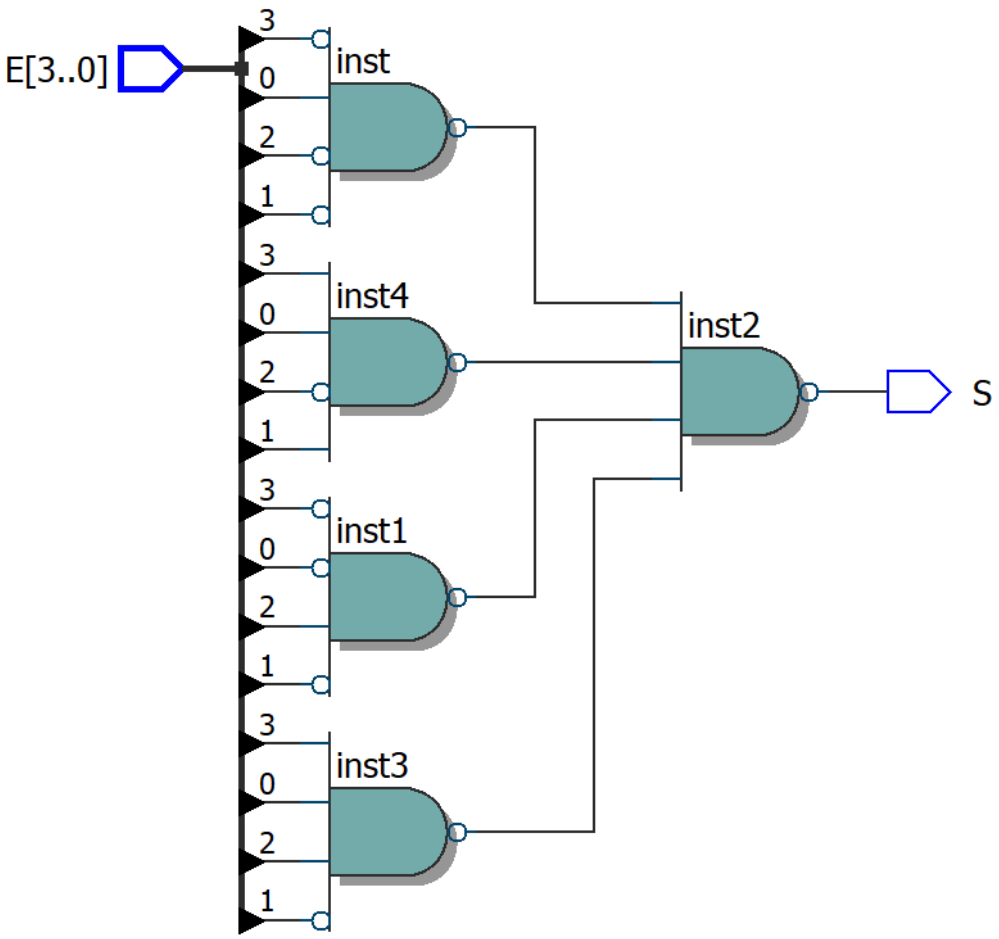
ÁREA: 37

1	4.457	A[2]	flag_N	clk	clk	10.000	0.000	5.143
2	4.449	B[2]	flag_N	clk	clk	10.000	0.000	5.151
3	4.211	op_sel[0]	flag_N	clk	clk	10.000	0.000	5.389
4	4.178	A[3]	flag_N	clk	clk	10.000	0.000	5.422
5	4.152	B[1]	flag_N	clk	clk	10.000	0.000	5.448
6	4.151	op_sel[1]	flag_N	clk	clk	10.000	0.000	5.449
7	4.021	A[0]	flag_N	clk	clk	10.000	0.000	5.579
8	4.016	B[3]	flag_N	clk	clk	10.000	0.000	5.584
9	4.010	A[3]	flag_Z	clk	clk	10.000	0.000	5.590
10	3.851	B[3]	flag_Z	clk	clk	10.000	0.000	5.749
11	3.849	B[0]	flag_N	clk	clk	10.000	0.000	5.751
12	3.832	A[1]	flag_N	clk	clk	10.000	0.000	5.768
13	3.821	A[0]	flag_Z	clk	clk	10.000	0.000	5.779
14	3.731	B[1]	flag_Z	clk	clk	10.000	0.000	5.869
15	3.669	op_sel[1]	flag_Z	clk	clk	10.000	0.000	5.931
16	3.660	B[2]	flag_Z	clk	clk	10.000	0.000	5.940
17	3.656	A[2]	flag_Z	clk	clk	10.000	0.000	5.944
18	3.649	B[0]	flag_Z	clk	clk	10.000	0.000	5.951
19	3.505	A[3]	S_b	clk	clk	10.000	0.000	6.095
20	3.498	op_sel[0]	flag_Z	clk	clk	10.000	0.000	6.102
21	3.496	B[2]	S_b	clk	clk	10.000	0.000	6.104
22	3.492	A[2]	S_b	clk	clk	10.000	0.000	6.108
23	3.467	B[2]	S_g	clk	clk	10.000	0.000	6.133
24	3.466	A[3]	S_g	clk	clk	10.000	0.000	6.134
25	3.463	A[2]	S_g	clk	clk	10.000	0.000	6.137
26	3.387	A[3]	S_d	clk	clk	10.000	0.000	6.213
27	3.385	A[1]	flag_Z	clk	clk	10.000	0.000	6.215
28	3.381	B[2]	S_d	clk	clk	10.000	0.000	6.219
29	3.377	A[2]	S_d	clk	clk	10.000	0.000	6.223
30	3.346	B[3]	S_b	clk	clk	10.000	0.000	6.254
31	3.335	A[0]	S_b	clk	clk	10.000	0.000	6.265
32	3.324	B[1]	S_b	clk	clk	10.000	0.000	6.276
33	3.307	B[3]	S_g	clk	clk	10.000	0.000	6.293
34	3.306	B[1]	S_g	clk	clk	10.000	0.000	6.294
35	3.296	A[0]	S_g	clk	clk	10.000	0.000	6.304
36	3.275	B[2]	S_f	clk	clk	10.000	0.000	6.325
37	3.271	A[2]	S_f	clk	clk	10.000	0.000	6.329
38	3.262	op_sel[1]	S_b	clk	clk	10.000	0.000	6.338
39	3.255	A[3]	S_f	clk	clk	10.000	0.000	6.345

40	3.244	op_sel[1]	S_g	clk	clk	10.000	0.000	6.356
41	3.228	B[3]	S_d	clk	clk	10.000	0.000	6.372
42	3.221	B[2]	S_c	clk	clk	10.000	0.000	6.379
43	3.218	A[3]	S_c	clk	clk	10.000	0.000	6.382
44	3.217	A[0]	S_d	clk	clk	10.000	0.000	6.383
45	3.217	A[2]	S_c	clk	clk	10.000	0.000	6.383
46	3.208	B[1]	S_d	clk	clk	10.000	0.000	6.392
47	3.190	B[2]	S_e	clk	clk	10.000	0.000	6.410
48	3.187	A[3]	S_e	clk	clk	10.000	0.000	6.413
49	3.186	A[2]	S_e	clk	clk	10.000	0.000	6.414
50	3.163	B[0]	S_b	clk	clk	10.000	0.000	6.437
51	3.154	op_sel[0]	S_b	clk	clk	10.000	0.000	6.446
52	3.146	op_sel[1]	S_d	clk	clk	10.000	0.000	6.454
53	3.136	op_sel[0]	S_g	clk	clk	10.000	0.000	6.464
54	3.124	B[0]	S_g	clk	clk	10.000	0.000	6.476
55	3.120	A[3]	S_a	clk	clk	10.000	0.000	6.480
56	3.104	B[2]	S_a	clk	clk	10.000	0.000	6.496
57	3.100	A[2]	S_a	clk	clk	10.000	0.000	6.500
58	3.096	B[3]	S_f	clk	clk	10.000	0.000	6.504
59	3.094	B[1]	S_f	clk	clk	10.000	0.000	6.506
60	3.085	A[0]	S_f	clk	clk	10.000	0.000	6.515
61	3.059	B[3]	S_c	clk	clk	10.000	0.000	6.541
62	3.048	A[0]	S_c	clk	clk	10.000	0.000	6.552
63	3.045	B[0]	S_d	clk	clk	10.000	0.000	6.555
64	3.038	op_sel[0]	S_d	clk	clk	10.000	0.000	6.562
65	3.036	B[1]	S_c	clk	clk	10.000	0.000	6.564
66	3.036	B[1]	S_e	clk	clk	10.000	0.000	6.564
67	3.032	op_sel[1]	S_f	clk	clk	10.000	0.000	6.568
68	3.028	B[3]	S_e	clk	clk	10.000	0.000	6.572
69	3.017	A[0]	S_e	clk	clk	10.000	0.000	6.583
70	2.978	A[1]	S_b	clk	clk	10.000	0.000	6.622
71	2.974	op_sel[1]	S_c	clk	clk	10.000	0.000	6.626
72	2.974	op_sel[1]	S_e	clk	clk	10.000	0.000	6.626
73	2.961	B[3]	S_a	clk	clk	10.000	0.000	6.639
74	2.960	A[1]	S_g	clk	clk	10.000	0.000	6.640
75	2.950	A[0]	S_a	clk	clk	10.000	0.000	6.650
76	2.924	B[1]	S_a	clk	clk	10.000	0.000	6.676
77	2.924	op_sel[0]	S_f	clk	clk	10.000	0.000	6.676
78	2.913	B[0]	S_f	clk	clk	10.000	0.000	6.687
79	2.876	B[0]	S_c	clk	clk	10.000	0.000	6.724
80	2.866	op_sel[0]	S_c	clk	clk	10.000	0.000	6.734
81	2.866	op_sel[0]	S_e	clk	clk	10.000	0.000	6.734
82	2.862	A[1]	S_d	clk	clk	10.000	0.000	6.738
83	2.862	op_sel[1]	S_a	clk	clk	10.000	0.000	6.738
84	2.845	B[0]	S_e	clk	clk	10.000	0.000	6.755
85	2.778	B[0]	S_a	clk	clk	10.000	0.000	6.822
86	2.754	op_sel[0]	S_a	clk	clk	10.000	0.000	6.846
87	2.748	A[1]	S_f	clk	clk	10.000	0.000	6.852
88	2.690	A[1]	S_c	clk	clk	10.000	0.000	6.910
89	2.690	A[1]	S_e	clk	clk	10.000	0.000	6.910
90	2.578	A[1]	S_a	clk	clk	10.000	0.000	7.022


Name
Value at 0 ps
E
B 0000
S_A
B X
S_B
B X
S_C
B X
S_D
B X
S_E
B X
S_F
B X
S_G
B X
0 ps
80.0 ns
160.0 ns
240.0 ns
320.0 ns
400.0 ns
480.0 ns
560.0 ns
640.0 ns
720.0 ns
800.0 ns
880.0 ns
960.0 ns
0000
0001
0010
0011
0100
0101
0110
0111
1000
1001
1010
1011
1100
1101
1110
1111
0000

FUNÇÕES DO DISPLAY:
SAÍDA A:

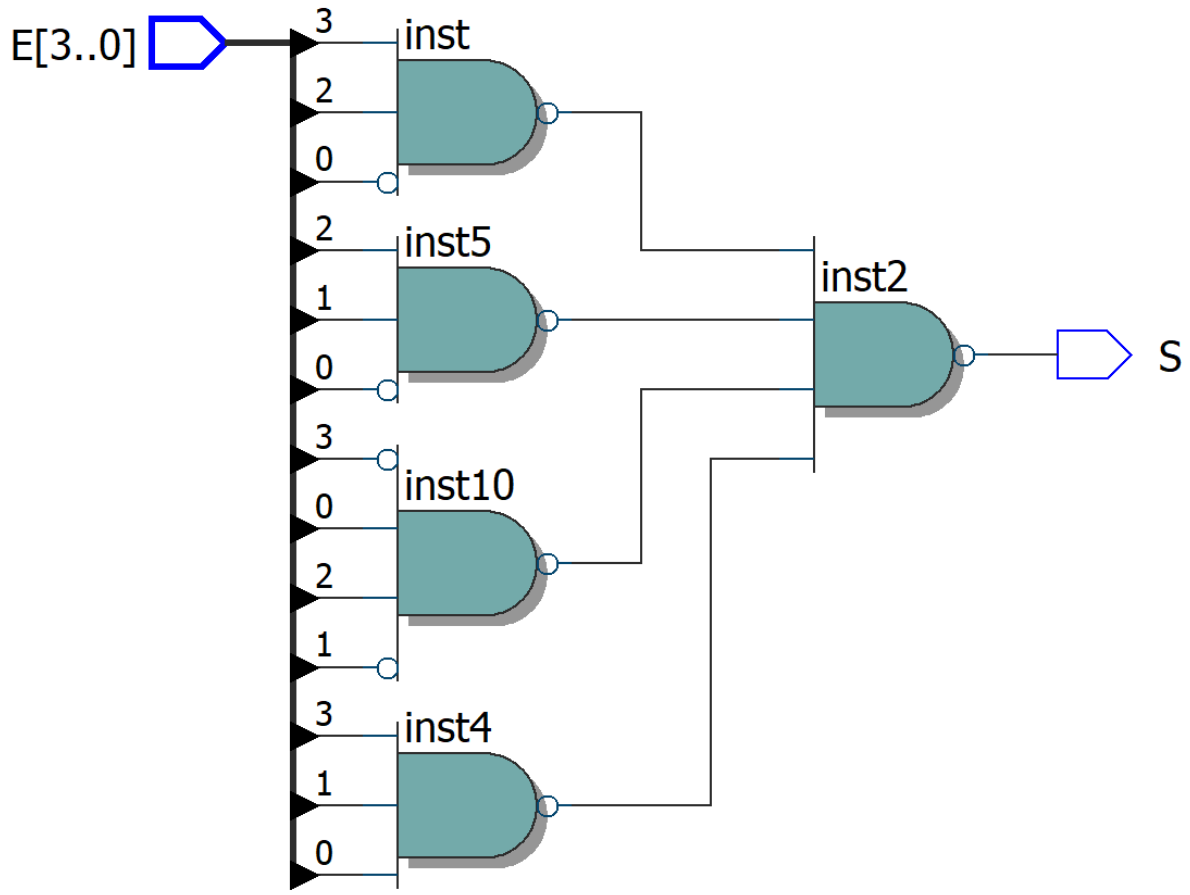


Flow Status	Successful - Thu May 15 20:33:56 2025
Quartus II 64-Bit Version	13.1.0 Build 162 10/23/2013 SJ Web Edition
Revision Name	ULA
Top-level Entity Name	S_A
Family	Cyclone III
Device	EP3C16F484C6
Timing Models	Final
Total logic elements	1 / 15,408 (< 1 %)
Total combinational functions	1 / 15,408 (< 1 %)
Dedicated logic registers	0 / 15,408 (0 %)
Total registers	0
Total pins	5 / 347 (1 %)
Total virtual pins	0
Total memory bits	0 / 516,096 (0 %)
Embedded Multiplier 9-bit elements	0 / 112 (0 %)
Total PLLs	0 / 4 (0 %)

ÁREA: 6

S_A:(!A*!B*!C*D)+(!A*B*!C*!D)+(A*B*!C*D)+(A*!B*C*D)

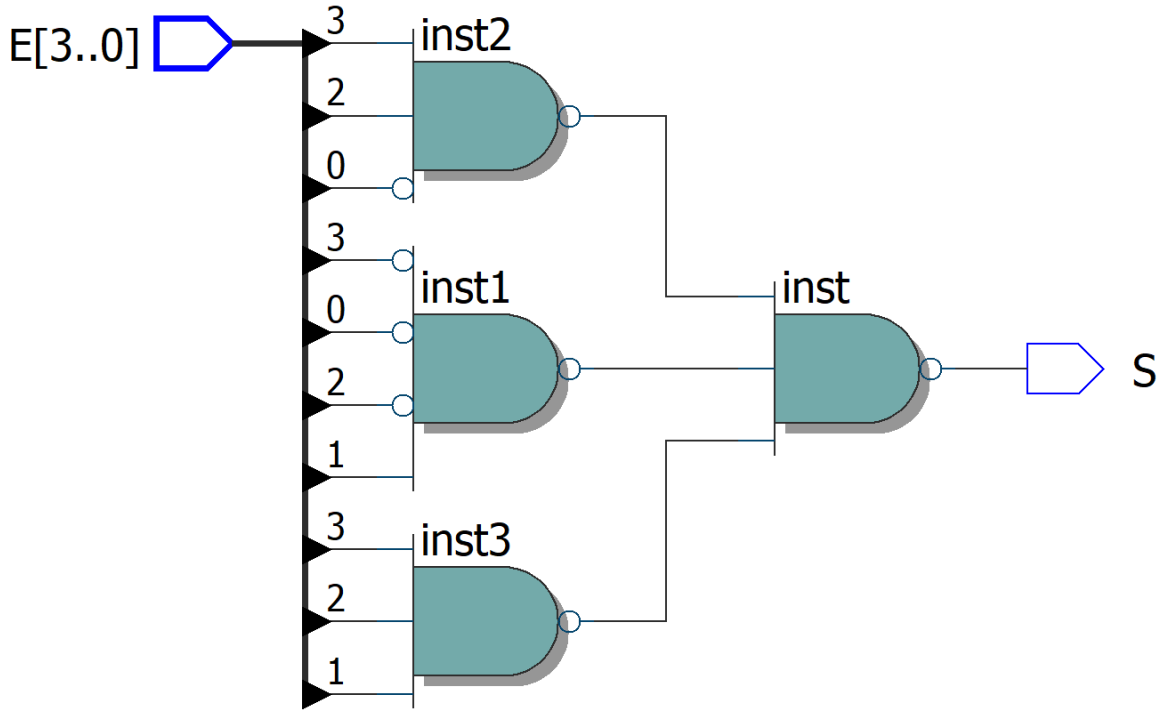
SAÍDA B:



Flow Summary	
Flow Status	Successful - Thu May 15 20:35:31 2025
Quartus II 64-Bit Version	13.1.0 Build 162 10/23/2013 SJ Web Edition
Revision Name	ULA
Top-level Entity Name	S_B
Family	Cyclone III
Device	EP3C16F484C6
Timing Models	Final
Total logic elements	1 / 15,408 (< 1 %)
Total combinational functions	1 / 15,408 (< 1 %)
Dedicated logic registers	0 / 15,408 (0 %)
Total registers	0
Total pins	5 / 347 (1 %)
Total virtual pins	0
Total memory bits	0 / 516,096 (0 %)
Embedded Multiplier 9-bit elements	0 / 112 (0 %)
Total PLLs	0 / 4 (0 %)

S_B:
(A*B*!D)+(!A*B*!C*D)+(A*C*D)+(B*C*!D)
ÁREA: 6

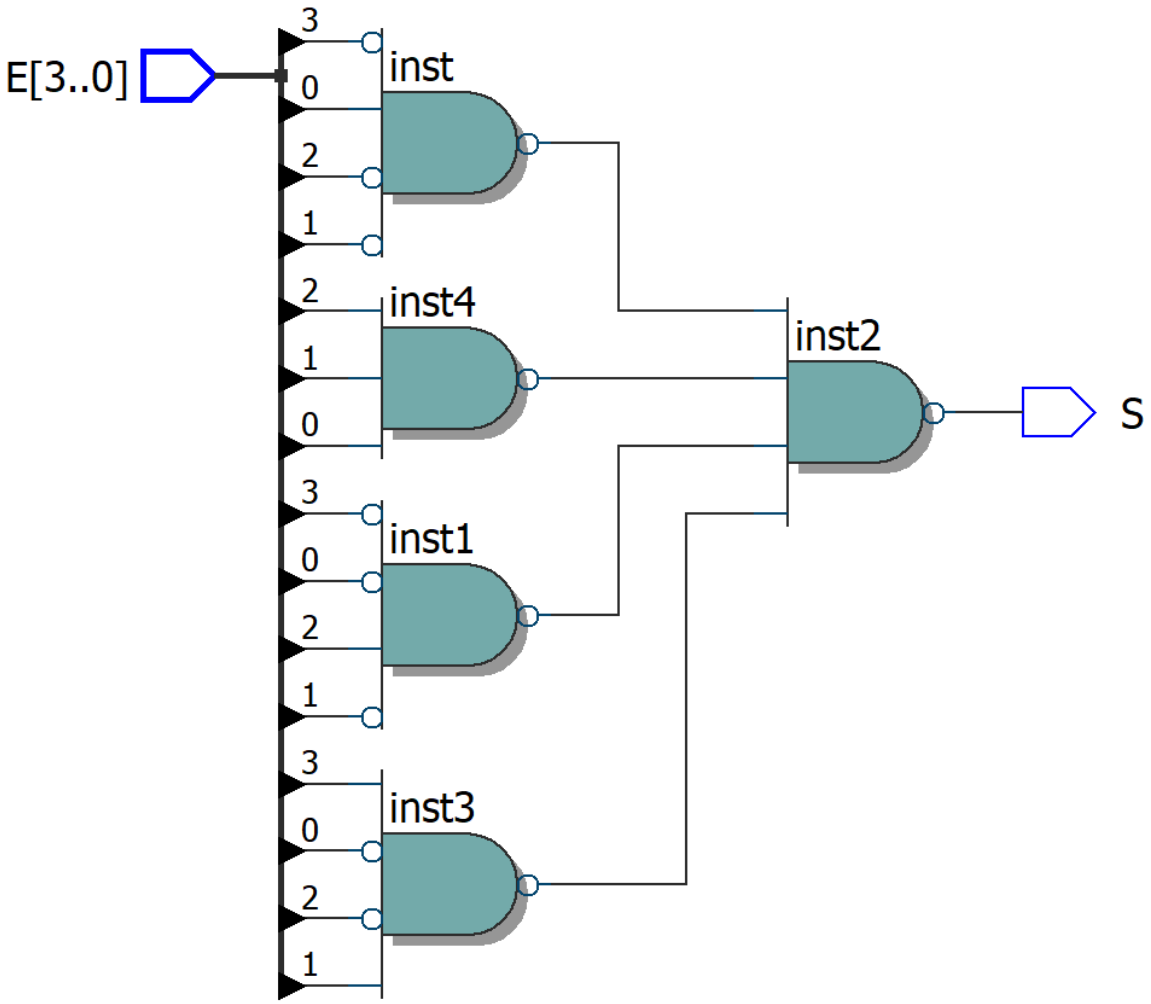
SAÍDA C:



Flow Summary	
Flow Status	Successful - Thu May 15 20:36:45 2025
Quartus II 64-Bit Version	13.1.0 Build 162 10/23/2013 SJ Web Edition
Revision Name	ULA
Top-level Entity Name	S_C
Family	Cyclone III
Device	EP3C16F484C6
Timing Models	Final
Total logic elements	1 / 15,408 (< 1 %)
Total combinational functions	1 / 15,408 (< 1 %)
Dedicated logic registers	0 / 15,408 (0 %)
Total registers	0
Total pins	5 / 347 (1 %)
Total virtual pins	0
Total memory bits	0 / 516,096 (0 %)
Embedded Multiplier 9-bit elements	0 / 112 (0 %)
Total PLLs	0 / 4 (0 %)

S_C:
(A*B*!D)+(!A*!B*C*!D)+(A*B*C)
ÁREA: 6

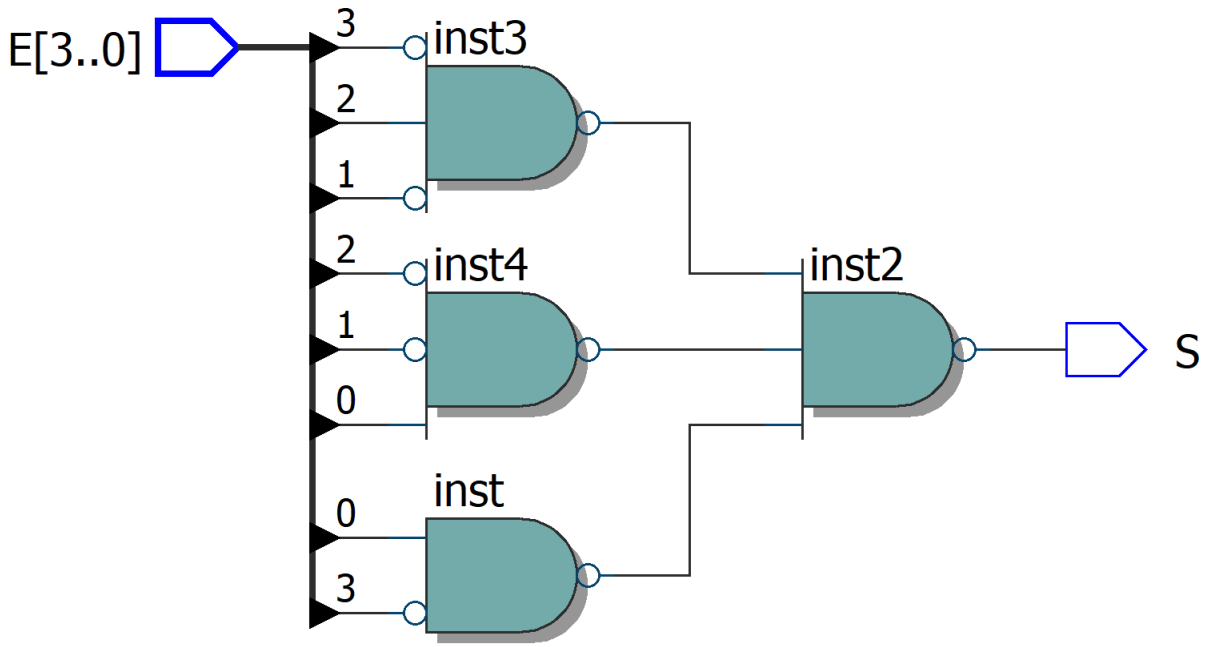
SAÍDA D:



Flow Summary	
Flow Status	Successful - Thu May 15 20:38:17 2025
Quartus II 64-Bit Version	13.1.0 Build 162 10/23/2013 SJ Web Edition
Revision Name	ULA
Top-level Entity Name	S_D
Family	Cyclone III
Device	EP3C16F484C6
Timing Models	Final
Total logic elements	1 / 15,408 (< 1 %)
Total combinational functions	1 / 15,408 (< 1 %)
Dedicated logic registers	0 / 15,408 (0 %)
Total registers	0
Total pins	5 / 347 (1 %)
Total virtual pins	0
Total memory bits	0 / 516,096 (0 %)
Embedded Multiplier 9-bit elements	0 / 112 (0 %)
Total PLLs	0 / 4 (0 %)

S_D:
(!A*!B*!C*D)+(!A*B*!C*!D)+(A*!B*C*!D)+(B*C*D)
ÁREA: 6

SAÍDA E:

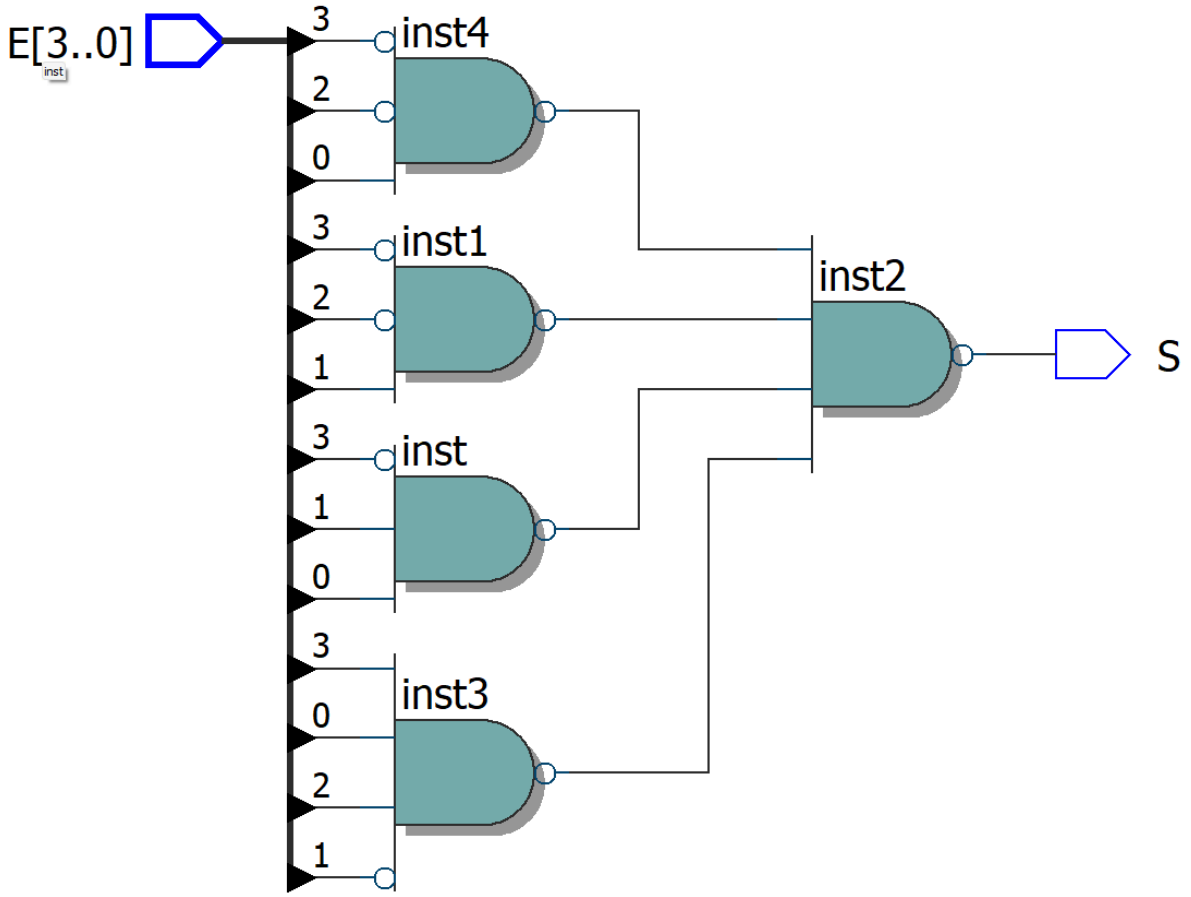


Flow Summary	
Flow Status	Successful - Thu May 15 20:39:22 2025
Quartus II 64-Bit Version	13.1.0 Build 162 10/23/2013 SJ Web Edition
Revision Name	ULA
Top-level Entity Name	S_E
Family	Cyclone III
Device	EP3C16F484C6
Timing Models	Final
Total logic elements	1 / 15,408 (< 1 %)
Total combinational functions	1 / 15,408 (< 1 %)
Dedicated logic registers	0 / 15,408 (0 %)
Total registers	0
Total pins	5 / 347 (1 %)
Total virtual pins	0
Total memory bits	0 / 516,096 (0 %)
Embedded Multiplier 9-bit elements	0 / 112 (0 %)
Total PLLs	0 / 4 (0 %)

S_E:
(!A*B*!C)+(!B*!C*D)+(!A*D)

ÁREA: 6

SAÍDA F:

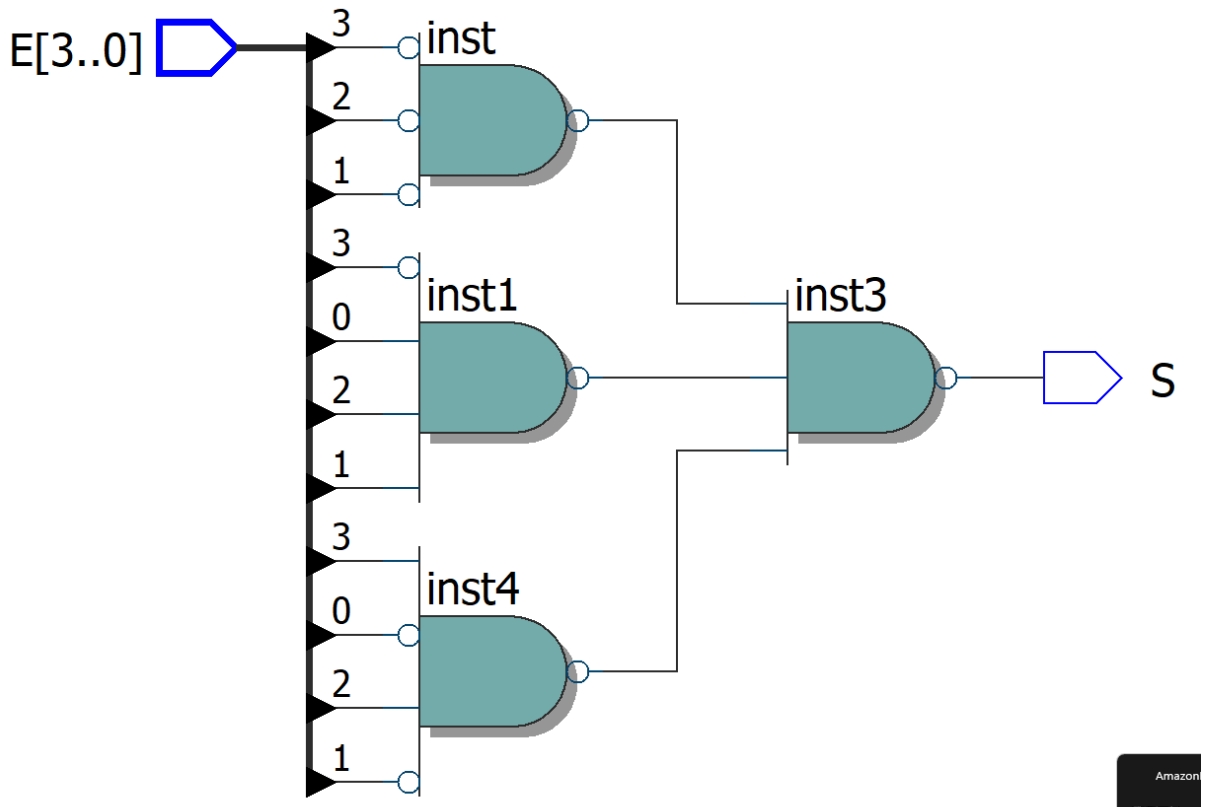


Flow Summary	
Flow Status	Successful - Thu May 15 20:40:25 2025
Quartus II 64-Bit Version	13.1.0 Build 162 10/23/2013 SJ Web Edition
Revision Name	ULA
Top-level Entity Name	S_F
Family	Cyclone III
Device	EP3C16F484C6
Timing Models	Final
Total logic elements	1 / 15,408 (< 1 %)
Total combinational functions	1 / 15,408 (< 1 %)
Dedicated logic registers	0 / 15,408 (0 %)
Total registers	0
Total pins	5 / 347 (1 %)
Total virtual pins	0
Total memory bits	0 / 516,096 (0 %)
Embedded Multiplier 9-bit elements	0 / 112 (0 %)
Total PLLs	0 / 4 (0 %)

S_F:
(!A*!B*D)+(!A*C*D)+(A*B*!C*D)+(!A*!B*C)

ÁREA: 6

SAÍDA G:

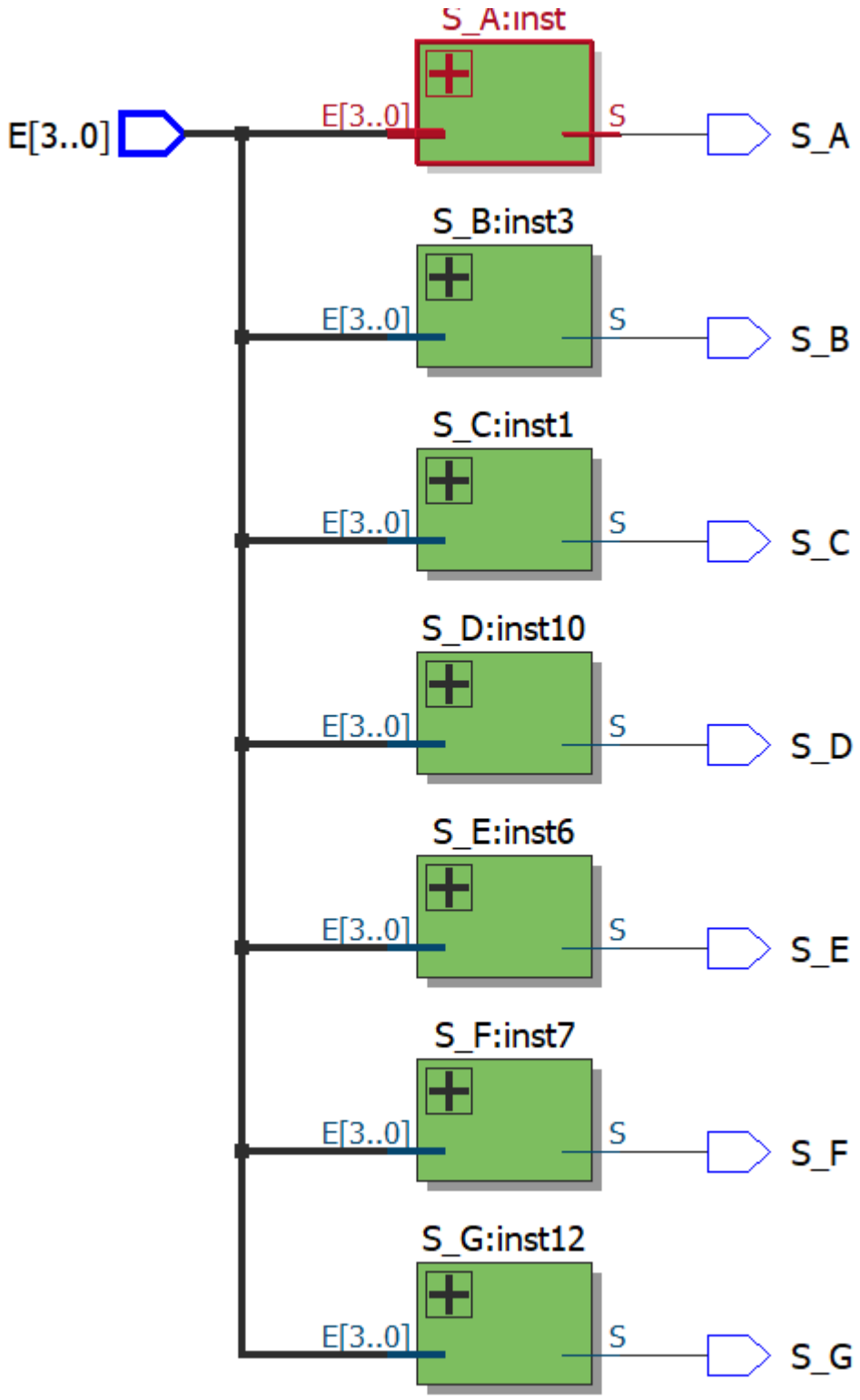


Flow Summary	
Flow Status	Successful - Thu May 15 20:41:22 2025
Quartus II 64-Bit Version	13.1.0 Build 162 10/23/2013 SJ Web Edition
Revision Name	ULA
Top-level Entity Name	S_G
Family	Cyclone III
Device	EP3C16F484C6
Timing Models	Final
Total logic elements	1 / 15,408 (< 1 %)
Total combinational functions	1 / 15,408 (< 1 %)
Dedicated logic registers	0 / 15,408 (0 %)
Total registers	0
Total pins	5 / 347 (1 %)
Total virtual pins	0
Total memory bits	0 / 516,096 (0 %)
Embedded Multiplier 9-bit elements	0 / 112 (0 %)
Total PLLs	0 / 4 (0 %)

S_G:
(!A*!B*!C)+(!A*B*C*D)+(A*B*!C*!D)
ÁREA: 6

DECODIFICADOR DISPLAY:

O decodificador incorpora todas as saídas mencionadas anteriormente.



Flow Summary	
Flow Status	Successful - Thu May 15 20:42:43 2025
Quartus II 64-Bit Version	13.1.0 Build 162 10/23/2013 SJ Web Edition
Revision Name	ULA
Top-level Entity Name	Decodificador_Display_7
Family	Cyclone III
Device	EP3C16F484C6
Timing Models	Final
Total logic elements	7 / 15,408 (< 1 %)
Total combinational functions	7 / 15,408 (< 1 %)
Dedicated logic registers	0 / 15,408 (0 %)
Total registers	0
Total pins	11 / 347 (3 %)
Total virtual pins	0
Total memory bits	0 / 516,096 (0 %)
Embedded Multiplier 9-bit elements	0 / 112 (0 %)
Total PLLs	0 / 4 (0 %)

ÁREA: 18